



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0003



Agricultural Machinery – Centrifugal Pumpset FARM ROVER FR20W

Test Report Number	2023 - 0003
Date of Issuance	March 3, 2023
Valid Until	March 3, 2028
Agricultural Machinery	Centrifugal Pumpset
Type	Single-end suction, 50 mm × 50 mm, Self-priming, Close-coupled
Brand	FARM ROVER
Model	FR20W
Test Requested by	TARLAC ROBERT "G" MOTORS TRADING #288 F. Tanedo St., Brgy. San Nicolas, Tarlac City

SUMMARY

Performance Criteria	Manufacturer's Specification	AMTEC Test
Discharge Capacity, L/s, maximum	10.00 ^a	9.26
Total Head, m, maximum	26 ^a	25.52
Suction Lift, m, maximum	7 ^a	8.13
System Efficiency, %, maximum	ND ^b	6.83
Fuel Consumption Rate, L/h, maximum	ND ^b	1.80

^a The data were obtained from the nameplate of the centrifugal pump.

^b Not specified by the requesting party.

Note: ND – No Data

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Not provided
d. Accessories and attachments	Not provided
e. Maintenance instruction	Provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly and installation	Provided
h. Specifications	Not provided
i. Dismantling and disposal	Not provided
j. Warranty	Not provided
k. Parts list	Not provided
2. Previous test reports of similar brand and model	None
3. Pumpset manufacturer	Not provided
4. Manufacturer's performance curve	Not provided
5. Prime mover manufacturer	HONDA POWER PRODUCTS CO. LTD. China
6. Safety features	None
7. The pump shall be designed for easy maintenance. Major parts such as casing components and bearing stoppings shall have accurate alignment of assembly.	The pump can be easily accessed for maintenance. Tools are needed to dismantle the assembly. The major parts have accurate alignment of assembly.
8. The rotating parts shall be dynamically balanced.	The dynamic balance of rotating parts was not verified.
9. The pump shall be designed to permit removal of the impeller, shaft seal, and bearing assembly without disturbing the inlet and outlet flange connection.	The inlet and outlet flange connections are integrated to the pump cover/casing. The casing had to be removed completely to access the impeller.
10. Radial bearings shall be of standard available design	Not applicable
11. Sealed type bearing shall be used, or the bearing housing shall be sealed.	Not applicable
12. Bearings and seals can be easily replaced without disturbing pump drive and mounting	The pumpset uses a mechanical seal which cannot be replaced without dismantling the casing and impeller.

OBSERVATIONS (Cont'n)

13. Shafts shall be of ample size and appropriate stiffness of to transmit prime mover rated power, ensure satisfactory packing or seal performance, minimize wear and risk of seizure, and take due consideration of static and dynamic thrust.	The shaft size and stiffness were not verified.
14. Castings shall be free of shrink holes, blowholes, cracks scale, blisters, and other similar injurious defects.	The castings are free from shrink holes, blowholes, cracks scale, blisters, and other similar injurious defects during the test operation.
15. Shaft shall be machined and properly finished throughout its length.	The shaft was not verified.
16. Other remarks	None

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	January 13, 2023
Location of Test	AMTEC Test Laboratory, AMTEC Building, Pili Drive, UPLB, College, Laguna
Items Tested	Operator's Manual, Total Head, Input Power, System Efficiency, Discharge Capacity, Fuel Consumption Rate, Water Power, Pump Shaft Speed, and Suction Condition

METHODS OF TEST

The centrifugal pumpset was tested based on the PAES 115:2000 Agricultural Machinery – Centrifugal, Mixed-Flow and Axial-Flow Water Pumps – Methods of Test.

DESCRIPTION OF THE MACHINE

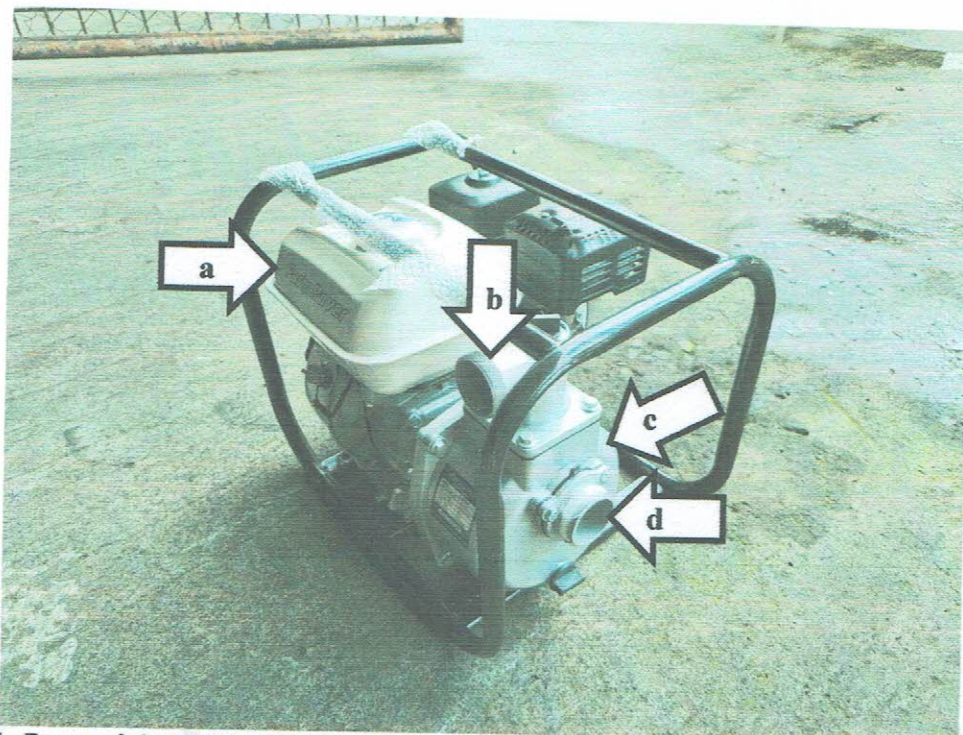


Figure 1. Parts of the FARM ROVER FR20W Centrifugal Pumpset include the following:
(a) Prime mover, (b) Discharge side, (c) Centrifugal pump, and (d) Suction side.

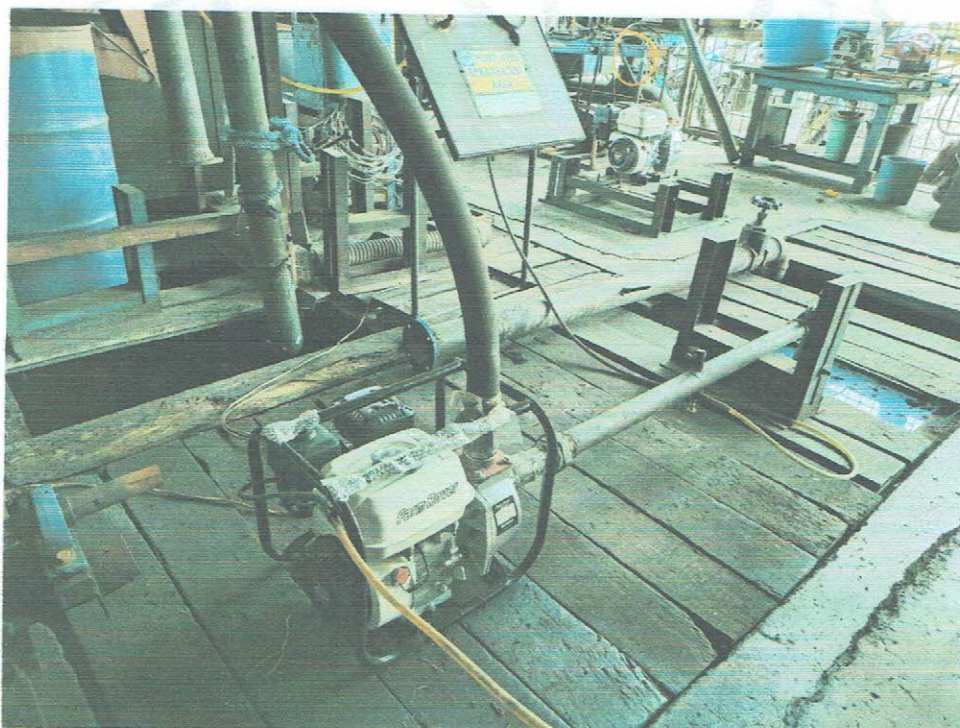


Figure 2. Pump test set-up of the of the FARM ROVER FR20W Centrifugal Pumpset.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Pumpset		
1.1	Pumpset overall dimensions, mm		
1.1.1	Length	ND	465
1.1.2	Width	ND	383
1.1.3	Height	ND	389
1.2	Weight, kg	ND	22
1.3	Power transmission system	ND	Closed-coupled
2	Pump		
2.1	Type	ND	Single-end suction, self-priming centrifugal pump
2.2	Pipe size, mm	50 × 50	50 × 50
2.3	Pump overall dimensions, mm		
2.3.1	Length	ND	210
2.3.2	Width	ND	205
2.3.3	Height	ND	327
2.4	Weight, kg	ND	NM ^b
2.5	Impeller		
2.5.1	Type	ND	Semi-open
2.5.2	Diameter, mm	ND	111.31
2.5.3	Width, mm	ND	23.16
2.5.4	Number of vanes	ND	3
2.5.5	Material	ND	S-5 ^c
2.6	Suction side		
2.6.1	Type	ND	Threaded
2.6.2	Diameter, mm	ND	50
2.6.3	Material	ND	Aluminum ^d
2.7	Discharge side		
2.7.1	Type	ND	Threaded
2.7.2	Diameter, mm	ND	50
2.7.3	Material	ND	Aluminum ^d

^a The requesting party did not provide complete machine specifications. Available data were obtained from the nameplate of the pumpset.

^b The pump is close-coupled with the prime mover and was not weighed separately.

^c Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer. See Table 5 for elemental composition.

^d Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer taken at a point in the pump casing. See Table 6 for elemental composition.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
2.8	Casing		
2.8.1	Type	ND	Volute
2.8.2	Material	ND	Aluminum ^b
2.9	Stuffing box	ND	Mechanical seal
2.10	Rotation	ND	Counterclockwise when facing the suction inlet
2.11	Maximum shaft speed, rpm	3600	3869
2.12	Maximum discharge, L/s	10	9.26
2.13	Maximum total head, m	26	25.52
2.14	Rated power, kW	ND	17.38 ^c
3	Prime mover		
3.1	Brand	HONDA	HONDA
3.2	Model	GP200	GP200
3.3	Make or manufacturer	ND	HONDA PRODUCTS CO. LTD.
3.4	Type	ND	Four stroke cycle, single inclined cylinder, OHV
3.5	Serial number	GCATH-1038802	GCATH-1038802
3.6	Rated power, kW	ND	ND ^d
3.6.1	Maximum	ND	ND ^d
3.6.2	Continuous	ND	ND ^d
3.7	Rated speed, rpm	ND	ND ^d
3.8	Displacement, cm ³	ND	ND ^d
3.9	Fuel type	ND	Gasoline
3.10	Fuel tank capacity, L	ND	3.57
3.11	Cooling system	ND	Air-cooled
3.12	Starting system	ND	Rope-recoil starter
3.13	Weight, kg	ND	NM ^e

^a The requesting party did not provide complete machine specifications. Available data were obtained from the nameplate of the pumpset.

^b Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer. See Table 6 for elemental composition.

^c Maximum input power observed at 0 L/s during the performance test.

^d Prime mover nameplate was not available at the time of test.

^e The prime mover is close-coupled with the pump and was not weighed separately.

Note: ND – No Data, NM – Not Measured

RESULTS

Performance Test

Table 1. Condition during the performance test of the FARM ROVER FR20W Centrifugal Pumpset.

Temperature, °C	27.2
Relative Humidity, %	85.0
Atmospheric Pressure, mb	1013.0
Pumping Water Temperature, °C	29.8

Table 2. Performance test data of the FARM ROVER FR20W Centrifugal Pumpset.

Pump Speed, rpm	Fuel Consumption, L/h	Total Head, m	Discharge Capacity, L/s	Water Power, kW	Input Power, kW	Pump Efficiency, %
3869	1.80	25.52	0	0	17.38	0
3854	1.77	22.62	1.53	0.339	17.15	1.98
3842	1.66	20.92	2.71	0.557	16.02	3.47
3811	1.59	19.35	4.17	0.792	15.36	5.16
3783	1.41	17.85	4.95	0.866	13.69	6.33
3783	1.49	15.91	6.02	0.939	14.45	6.50
3785	1.55	14.48	7.08	1.00	15.02	6.69
3787	1.47	12.98	7.66	0.974	14.25	6.83
3795	1.37	9.84	8.85	0.854	13.24	6.45
3805	1.42	8.20	9.26	0.745	13.74	5.42

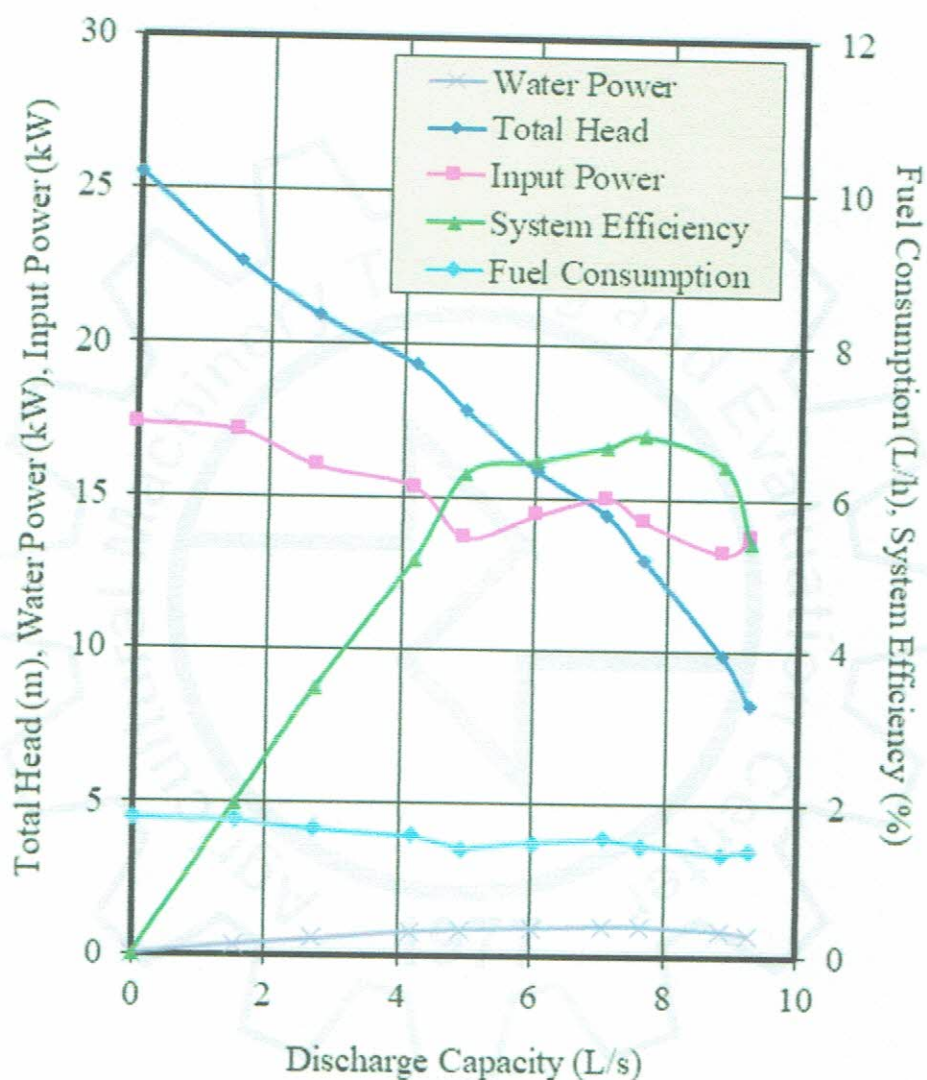
RESULTS (Cont'n)

Figure 3. Performance characteristic curves of the FARM ROVER FR20W Centrifugal Pumpset.

RESULTS (Cont'n)**Cavitation Test**

Table 3. Condition during the cavitation test of the FARM ROVER FR20W Centrifugal Pumpset.

Temperature, °C	27.5
Relative Humidity, %	84.4
Atmospheric Pressure, mb	1013.0
Pumping Water Temperature, °C	29.7

Table 4. Cavitation test data of the FARM ROVER FR20W Centrifugal Pumpset.

Pump Shaft Speed, m	Total Suction Lift, m	Net Positive Suction Head Available, m	Total Head, m	Discharge Capacity, L/s
3771	4.36	5.56	8.41	9.20
3790	5.33	4.59	9.09	7.85
3813	6.24	3.68	9.77	6.54
3846	7.21	2.70	10.45	4.37
3922	8.13	1.79	11.13	0

RESULTS (Cont'n)

Table 5. Elemental composition of the impeller of the FARM ROVER FR20W Centrifugal Pumpset.

Element	Percent Composition^a
Aluminum	<LOD
Silicon	2.79
Phosphorus	<LOD
Sulfur	<LOD
Titanium	0.07
Vanadium	0.07
Chromium	0.07
Manganese	0.51
Iron	96.14
Cobalt	<LOD
Nickel	0.03
Copper	0.13
Niobium	<LOD
Molybdenum	0.08
Tin	0.06
Tungsten	<LOD
Lead	<LOD

^a Determined using BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer.

Note: LOD – Limit of Detection

RESULTS (Cont'n)

Table 6. Elemental composition of the pump casing of the FARM ROVER FR20W Centrifugal Pumpset.

Element	Percent Composition^a
Magnesium	<LOD
Aluminum	43.82
Silicon	4.91
Phosphorus	0.27
Sulfur	5.72
Titanium	10.85
Vanadium	2.67
Chromium	0.14
Manganese	0.63
Iron	6.90
Cobalt	0.04
Nickel	0.27
Copper	6.07
Zinc	6.71
Arsenic	<LOD
Selenium	<LOD
Yttrium	<LOD
Zirconium	0.11
Niobium	<LOD
Molybdenum	0.24
Ruthenium	1.40
Palladium	0.42
Silver	<LOD
Cadmium	<LOD
Indium	1.90
Tin	1.52
Antimony	<LOD
Tellurium	0.34
Hafnium	<LOD
Tantalum	0.42
Tungsten	<LOD
Rhenium	<LOD
Iridium	<LOD
Platinum	<LOD
Gold	<LOD
Mercury	<LOD
Lead	4.64
Bismuth	<LOD

^a Determined using BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer.

Note: LOD – Limit of Detection

OBSERVATIONS

Figure 4. Nameplate of the FARM ROVER FR20W Centrifugal Pumpset.



Figure 5. Markings and labels of the FARM ROVER FR20W Centrifugal Pumpset.

Tested by:


KAREN CYBILL F. CABANGLAN

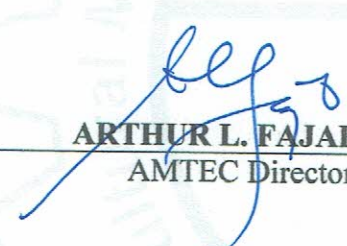
Test Engineer

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AMTEC Director

MAR 07 2023

Date

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